

الجامعة العربية الأمريكية
ARAB AMERICAN UNIVERSITY



SceneFroge

Procedural 2D and 3D
Course Project Report

Seed: 671

Modules:

- Shapes and Colour
- Sierpinski Fractal
- Transformations
- 3D Camera
- Measurements

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Introduction

SceneForge is a p5.js project that shows the basic 2D and 3D computer graphics concepts.

The project consists of Five modules covering shapes, recursion, transformations, cameras and measurements.

Seed and Scene Generation

I used seed 671 to generate a repeatable scene.

My ID: 202310671

The last 4 numbers are: 0671 so the seed is (671).

The seed produces 15 shapes, 6 colours and recursion depth 5.

```
seed = 671
```

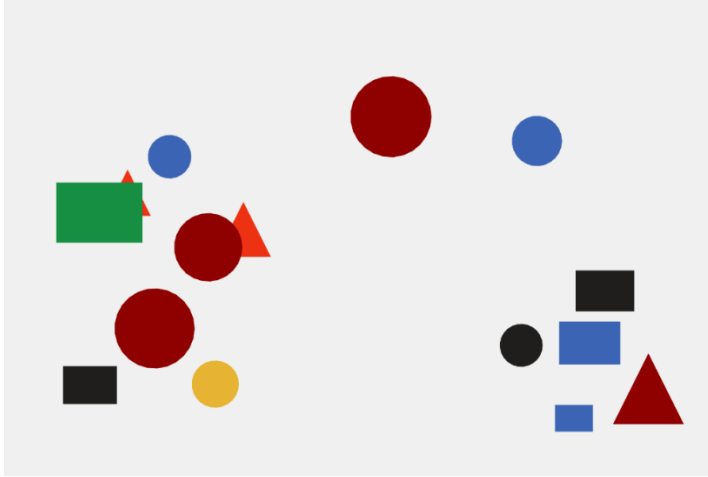
```
shapes = 15
```

```
palette = 6
```

```
depth = 5
```

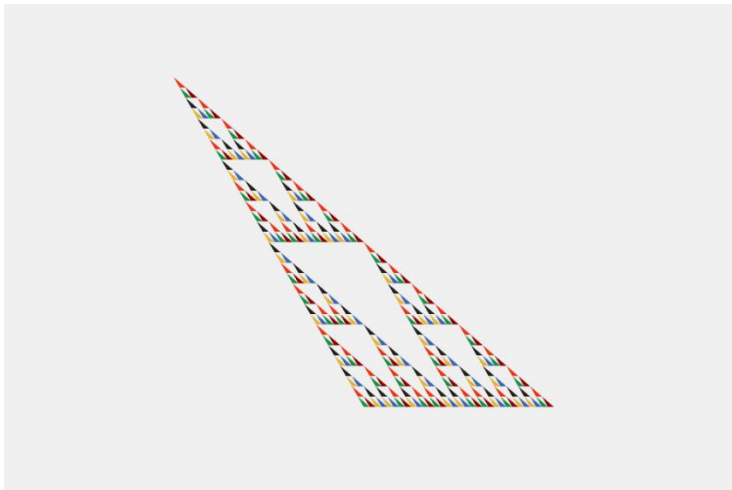
Module 1: Shapes and Colours

This module generates rectangles, circles and triangles using different positions, sizes and colours.



Module 2: Sierpinski

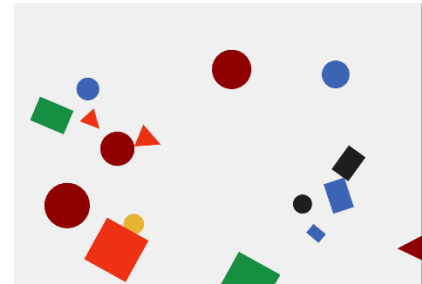
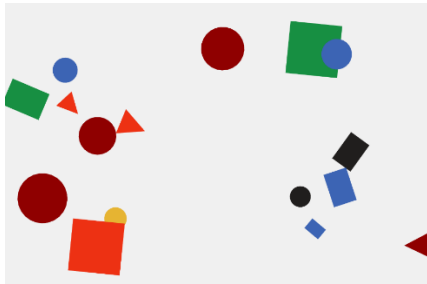
This module uses recursion to create Sierpinski triangle by dividing the triangle into 3 smaller ones with depth=5 so the number of triangles is $3^5 = 243$.



Module 3: Transformations

This module applies translation, rotation and scaling to the objects.

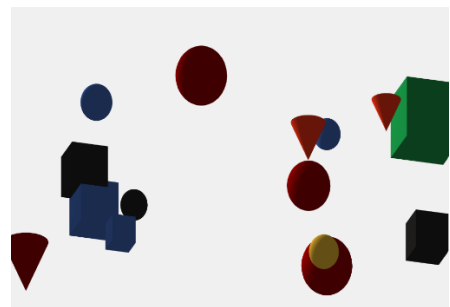
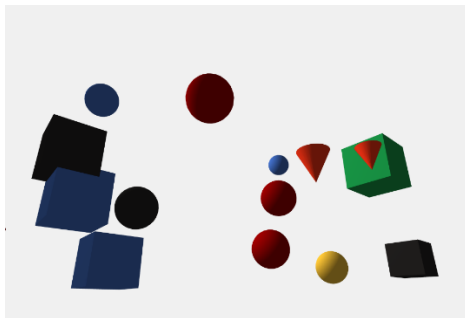
Each object uses the three transformations, $15 \times 3 = 45$ transformations.



Module 4: 3D Camera

This module displays 3D boxes using WebGL lighting and a moving camera.

It supports both perspective and orthographic using the (p) and (o) keys.



Module 5: Measurements

This module displays the number of shapes, triangles, transformations and recursion depth.

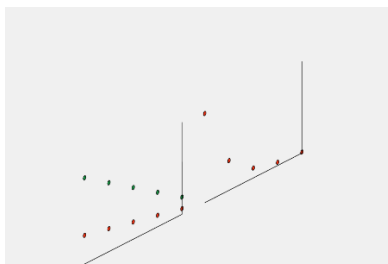
The measurements are based of the number of operations the program does.

It shows the relationship between input size transformations and fractal depth.

Measure	Basic operation	Operation count	Growth class
Number of shapes(n)	Draw one shape	$C(n) = n$	$O(n)$
Recursion depth(d)	Create 3 sub-triangles	$C(d)=3^d$	$O(3^d)$
Number of shapes(n)	Apply 3 transformation per shape	$C(n)=n$	$O(n)$
Recursion depth	One recursion level	$C(d)=d+1$	$O(d)$

Calculation:

- Shapes drawn per frame
Each shape is drawn once $C(n)=n$
For $n = 15$ $C(15) = 15$.
- Fractal triangles
Each triangle creates 3 smaller triangles $C(d)=3^d$
For $d=5$ $C(5) = 3^5=243$
- Transforms applied
each shape uses 3 transformations $C(n)=3*n$
for $n=15$ $C(n)= 3*15 = 45$.
- Fractal recursion depth
the recursion goes through one level at a time until the depth is 0
 $C(d)= d+1$
for $d= 5$ $C(5) = 5+1 = 6$ levels.



```

Console
Shape results = ▶ (5) [5, 10, 15, 20, 25]
Triangle results = ▶ (5) [3, 9, 27, 81, 243]
Transform results = ▶ (5) [15, 30, 45, 60, 75]
seed = 671
shapes = 15
palette = 6
depth = 5

```

Experimental Results

I tested the code with different numbers of shapes and a different fractal depth every test was runned 3 times and the average was calculated

Shapes and Transforms

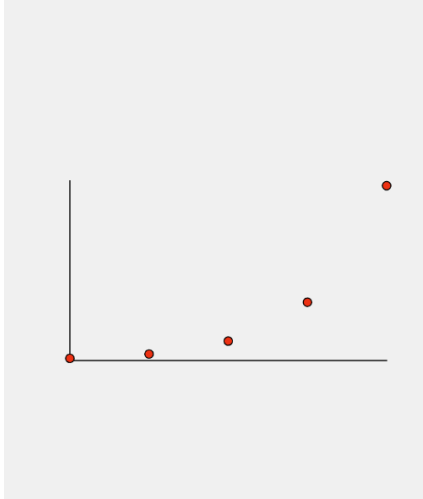
Number of shapes	Run 1	Run 2	Run 3	Average shapes	Average transforms
5	5	5	5	5	15
10	10	10	10	10	30
15	15	15	15	15	45
20	20	20	20	20	60
25	25	25	25	25	75

Fractal Results

Recursion depth	Run 1	Run 2	Run 3	Average
1	3	3	3	3
2	9	9	9	9
3	27	27	27	27
4	81	81	81	81
5	243	243	243	243

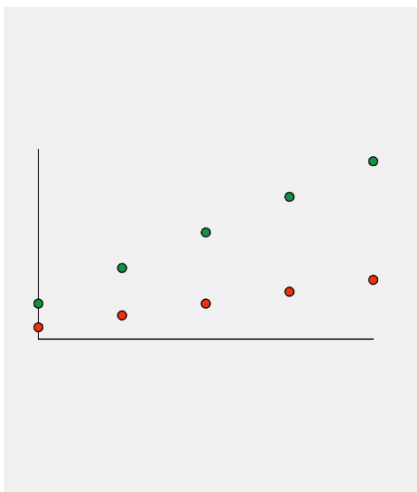
Graphs

Fractal Triangles



This screen shot it shows how the number of triangle increases as the recursion depth increases, the number of triangles grows as 3^d .

Shapes and Transformations



Here it shows the number of shapes and transformations for different input sizes. Both increase linearly as the number of shapes increases.

Testing

The project was tested, I ran all five modules, and the results were checked with the values that I expected.

The keyboard keys were also tested to switch between the modules and the camera views.

- 1 Shapes.
- 2 Fractal.
- 3 Transformations.
- 4 3D Camera.
- P Perspective Camera.
- O Orthographic Camera.
- 5 Measurements.

Conclusion

This project helped me to understand more about shapes, colours, transformations, recursion and how the 3D cameras work in p5.js.

The project showed the shapes and transformations grow linearly on the other hand the number of fractal triangles grows with the recursion depth.

AI Use Declaration

I used ai tools to help me understand some codes and the github commits, and to check whether my work was right or wrong.

The code and the final work were written and understood by me.